

4(a)(i)	human immunodeficiency virus ;	1
4(a)(ii)	<i>any four from:</i> there is a decrease in the percentage infections in most regions / AW ; <i>ref. to exceptions to this decreasing trend ; e.g. B (Eastern Europe and Central Asia) / D (Middle East and North Africa)</i> <i>allow no overall trend if mp 1 and mp 2 not awarded</i> the biggest percentage decrease, in E (East and Southern Africa) ; the smallest percentage decrease, in G (Latin America) ; data quote to support described trend ; AVP ;	4

Question	Answer	Marks
4(b)	any four from: tuberculosis / TB, is an opportunistic, disease / infection ; weakened immune system ; (fewer T-helper cells so a) less effective primary immune response ; (fewer helper T-cells so) less cytokine secreted ; less, phagocytosis of the pathogens / AW ; A ref. to angry macrophages fewer B-lymphocytes, stimulated / AW, to divide by mitosis and form plasma cells ; fewer antibodies secreted so fewer pathogens destroyed ; ref. to latent TB becoming activated ; AVP ; e.g. 'weak' / no, response to (BCG) vaccine so, fewer / no, memory cells TB is caused by, a bacterial pathogen / <i>Mycobacterium tuberculosis</i> / <i>Mycobacterium bovis</i>	4

Question	Answer	Marks
4(c)	any four from: 1 <i>ref.</i> to compression on ventricles ; 2 (so) volume of the ventricles reduces / ventricles cannot fully increase in size / AW ; 3 less blood, filling / leaving, ventricles (each cardiac cycle) ; 4 <i>idea that</i> ventricular, contraction / systole, less powerful / weaker / less forceful ; 5 less blood pumped to the, pulmonary / systemic, circulation (per, contraction / cardiac cycle / unit time ; A lungs / rest of body <i>faster breathing rate to compensate for</i> 6 consequence in, lungs / rest of body ; e.g. less oxygen, diffuses / AW, into the blood less carbon dioxide, diffuses / AW, out of the blood less oxygen delivered to the tissues / AW 7 faster breathing rate to (try and) increase quantity of oxygen, absorbed / AW (in the lungs) ; A increase quantity of carbon dioxide excreted 8 AVP ; e.g. heart / cardiac muscle, does not receive enough, glucose / oxygen	4

Question	Answer	Marks
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